1. recall that atoms of each element have different proton numbers;
2. understand that arranging the elements in order of their proton numbers gives repeating patterns in the properties of elements;
3. be able to use the Periodic Table to obtain the names, symbols, relative atomic masses and proton numbers of elements;
4. recall that a group of elements is a vertical column in the Periodic Table and that the elements have similar properties;
5. recall that a period is a row of elements in the Periodic Table;
6. be able to use the Periodic Table to classify an element as a metal or non-metal;
7. be able to use patterns in the Periodic Table to interpret data and predict properties of elements.

ⓘ Candidates will be given a copy of the Periodic Table (as in Appendix H) with the examination paper.

8. recall and recognise the chemical symbols for the group 1 metals: lithium, sodium and potassium;
9. recall that the alkali metals tarnish rapidly in moist air but are shiny when freshly cut;
10. be able to use qualitative and quantitative data to identify patterns and make predictions about the properties of group 1 metals (for example melting point, boiling point, density, formulae of compounds and relative reactivity);
11. describe the reactions of lithium, sodium and potassium with cold water;
12. recall that alkali metals react with water to form hydrogen and an alkaline solution of a hydroxide with the formula MOH;
13. recall that alkali metals react vigorously with chlorine to form colourless, crystalline salts with the formula MCl;
14. understand and be able to give examples to show that the alkali metals become more reactive as the group is descended;
15. recall the main hazard symbols and be able to give the safety precautions for handling hazardous chemicals (limited to harmful, toxic, irritant, corrosive, oxidizing, highly flammable);
16. explain the precautions necessary when working with group 1 metals and alkalis;
17. recall and recognise the chemical symbols for the atoms and molecules of the group 7 elements: chlorine, bromine and iodine;
18. recall the states of the halogens at room temperature and pressure;
19. recall the colours of the halogens in their normal physical state at room temperature and as gases;
20. recall that the halogens consist of diatomic molecules;
21. recall that the halogens can bleach dyes and kill bacteria in water;

22. be able to use qualitative and quantitative data to identify patterns and make predictions about the properties of the group 7 elements (for example melting point, boiling point, formulae of compounds and relative reactivity);

23. recall and be able to give examples to show that the halogens become less reactive as the group is descended;

24. explain the safety precautions necessary when working with the halogens;

25. recall the formulae of:
    - hydrogen, water and halogen molecules;
    - the halides and hydroxides of group 1 metals;

26. be able to balance unbalanced symbol equations;

27. be able to write balanced equations to describe the chemical reactions of group 1 metals with water and halogens;

28. recall and use state symbols: (s), (l), (g) and (aq) in equations.